

Efficiency–TFP Analysis

Usa

MAPS Project — Deliverable 4.2 (WP4: Improving Integrated Assessment Modelling)

Period: 1960–2019 • Observations: 60 • Date: 22 May 2026

This report documents the Efficiency–TFP cointegration analysis for Usa (1960–2019). Following De Ketelaere et al. (2026) and Santos et al. (2021), it tests whether changes in Total Factor Productivity (TFP) are explained by improvements in final-to-useful exergy efficiency, and assesses the stability of useful exergy intensity for scenario analysis.

Steps of the Analysis

- 1. Output elasticities (a_K, a_L) from income data
- 2. Labour inputs: unadjusted and quality-adjusted
- 3. TFP: unadjusted and quality-adjusted (z-normalised to base year)
- 4. Exergy efficiency and useful exergy intensity
- 5. Unit root tests (ADF and PP) on TFP and efficiency series
- 6. VAR estimation and residual diagnostics
- 7. Johansen cointegration tests: Hz (no deterministic) and Hc (restricted constant; no trend)
- 8. Useful exergy intensity constancy assessment

Key Results

i) Long-run TFP–Efficiency relationship [Hc specification — restricted constant; no trend]:

$$TFP = \exp(0.5132) \times EFF^{0.6377}$$
 (quality-adjusted TFP)

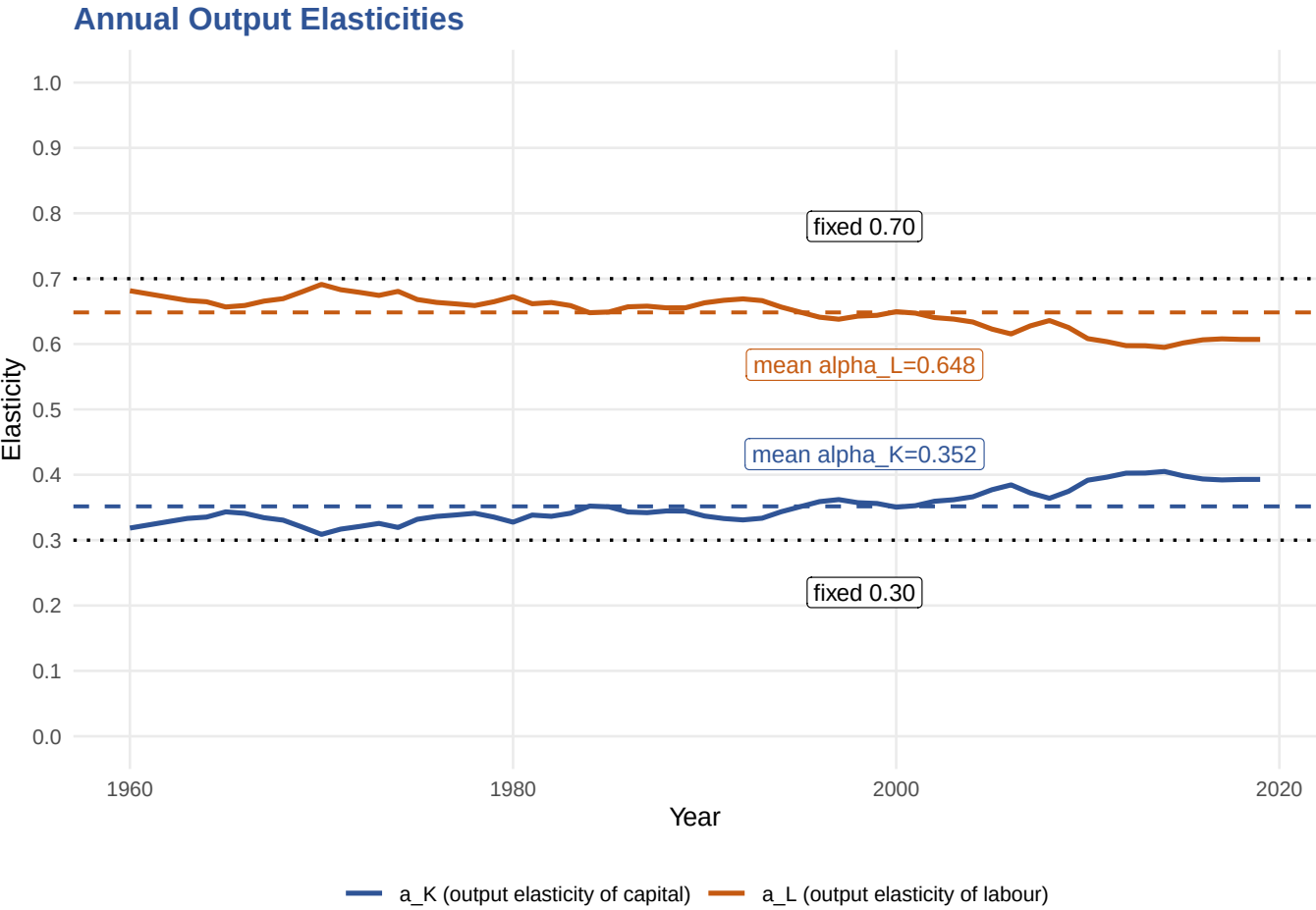
(z-form: $z_TFP = 0.6377 \times z_EFF + 0.5132$)

ii) Aggregate production function (Cobb–Douglas):

$$Y = \exp(0.5132) \times K_services^{0.3517} \times hL^{0.6483} \times EFF^{0.6377}$$

iii) Useful exergy intensity:

Not constant. Trend: –1.205% p.a. (R²=0.974)



Normalised TFP and Final-to-Useful Exergy Efficiency [$z = \ln(x/x_0) + 1$]



Unit Root Tests — ADF and PP (5% significance level)

Series: z_tfp_unadj | Verdict: I(1) | ADF: I(1) | PP: I(1)

Test	Specification	Statistic	CV 1%	CV 5%	CV 10%	Aux.	Val.	H0 rej.?	Verdict
ADF	Levels (trend+intercept)	-2.6591	-4.1211	-3.4425	-3.1070	Lags (BIC)	0	No	I(1)
ADF	1st diff. (intercept)	-6.3339	-3.3232	-2.7807	-2.5052	Lags (BIC)	0	Yes	—
PP	Levels (trend+intercept)	-2.7788	-4.1190	-3.4862	-3.1711	BW (NW)	3	No	I(1)
PP	1st diff. (intercept)	-6.3306	-3.5456	-2.9118	-2.5932	BW (NW)	3	Yes	—

Series: z_tfp_adj | Verdict: I(1) | ADF: I(1) | PP: I(1)

Test	Specification	Statistic	CV 1%	CV 5%	CV 10%	Aux.	Val.	H0 rej.?	Verdict
ADF	Levels (trend+intercept)	-1.8947	-4.1211	-3.4425	-3.1070	Lags (BIC)	0	No	I(1)
ADF	1st diff. (intercept)	-6.7657	-3.3232	-2.7807	-2.5052	Lags (BIC)	0	Yes	—
PP	Levels (trend+intercept)	-2.0737	-4.1190	-3.4862	-3.1711	BW (NW)	3	No	I(1)
PP	1st diff. (intercept)	-6.7458	-3.5456	-2.9118	-2.5932	BW (NW)	3	Yes	—

Series: z_eff | Verdict: I(1) | ADF: I(1) | PP: I(1)

Test	Specification	Statistic	CV 1%	CV 5%	CV 10%	Aux.	Val.	H0 rej.?	Verdict
ADF	Levels (trend+intercept)	-2.5302	-4.1211	-3.4425	-3.1070	Lags (BIC)	0	No	I(1)
ADF	1st diff. (intercept)	-8.3776	-3.3232	-2.7807	-2.5052	Lags (BIC)	0	Yes	—
PP	Levels (trend+intercept)	-2.4518	-4.1190	-3.4862	-3.1711	BW (NW)	3	No	I(1)
PP	1st diff. (intercept)	-8.4510	-3.5456	-2.9118	-2.5932	BW (NW)	3	Yes	—

[OK] All series are I(1): unit roots confirmed in levels, stationary in first differences.

Green light — Johansen cointegration analysis is appropriate.

VAR Residual Portmanteau Tests

VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Lags	Q-Stat	Prob.	Adj Q-Stat	Prob.	df
h=1	0.029828	---	0.030352	---	---
h=2	3.776166	---	3.910487	---	---
h=3	13.664248	0.0084	14.337919	0.0063	4
h=4	17.929581	0.0218	18.919202	0.0153	8
h=5	23.613909	0.0229	25.139788	0.0142	12

* Test is valid only for lags larger than the VAR lag order.

VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Lags	Q-Stat	Prob.	Adj Q-Stat	Prob.	df
h=1	0.023171	---	0.023577	---	---
h=2	3.602794	---	3.731044	---	---
h=3	14.363196	0.0062	15.078377	0.0045	4
h=4	17.780199	0.0229	18.748491	0.0163	8
h=5	24.353446	0.0182	25.941856	0.0109	12

* Test is valid only for lags larger than the VAR lag order.

[!!] WARNING: Significant autocorrelation detected. VAR residuals are not white noise — results may be unreliable.

VAR Residual Serial Correlation (LM) Tests

VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Null: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	1.153147	4	0.8858	0.278741	(4, 100.0)	0.8911
h=2	3.554867	4	0.4696	0.869308	(4, 100.0)	0.4852
h=3	10.864660	4	0.0281	2.752994	(4, 100.0)	0.0321
h=4	4.556098	4	0.3359	1.119560	(4, 100.0)	0.3516
h=5	6.581705	4	0.1597	1.633277	(4, 100.0)	0.1718

Null: No serial correlation at lags 1 to h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	1.153147	4	0.8858	0.278741	(4, 100.0)	0.8911
h=2	13.400688	8	0.0988	1.650272	(8, 96.0)	0.1208
h=3	16.935556	12	0.1520	1.355855	(12, 92.0)	0.2018
h=4	21.551938	16	0.1583	1.266461	(16, 88.0)	0.2369
h=5	30.158363	20	0.0673	1.412907	(20, 84.0)	0.1394

* Edgeworth expansion corrected likelihood ratio statistic.

VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Null: No serial correlation at lag h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	1.972195	4	0.7409	0.478609	(4, 100.0)	0.7514
h=2	3.436172	4	0.4876	0.839800	(4, 100.0)	0.5031
h=3	11.930481	4	0.0179	3.038876	(4, 100.0)	0.0207
h=4	3.867545	4	0.4242	0.947201	(4, 100.0)	0.4401
h=5	8.251689	4	0.0828	2.064394	(4, 100.0)	0.0911

Null: No serial correlation at lags 1 to h

Lag	LRE* stat	df	Prob.	Rao F–stat	df	Prob.
h=1	1.972195	4	0.7409	0.478609	(4, 100.0)	0.7514
h=2	8.011834	8	0.4323	0.960983	(8, 96.0)	0.4711
h=3	17.837315	12	0.1207	1.434427	(12, 92.0)	0.1647
h=4	22.521577	16	0.1271	1.329843	(16, 88.0)	0.1977
h=5	32.564791	20	0.0376	1.544297	(20, 84.0)	0.0879

* Edgeworth expansion corrected likelihood ratio statistic.

[!!] WARNING: Significant serial correlation detected in VAR residuals — consider increasing VAR lag order.

VAR Residual Normality Tests

VAR(2): TFP_unadj ~ EFF | T=58 (1962–2019)

Component	Skewness	Chi-sq	df	p	Kurtosis	Chi-sq	df	p	JB stat	df	p
Eq.1	−0.746650	5.389032	1	0.0203	4.052709	2.678140	1	0.1017	8.067172	2	0.0177
Eq.2	1.147339	12.725079	1	0.0004	9.758434	110.384695	1	0.0000	123.109774	2	0.0000

Test	Statistic	df	Prob
Joint skewness	18.114111	2	0.0001
Joint kurtosis	113.062835	2	0.0000
Joint JB	131.176946	4	0.0000

VAR(2): TFP_adj ~ EFF | T=58 (1962–2019)

Component	Skewness	Chi-sq	df	p	Kurtosis	Chi-sq	df	p	JB stat	df	p
Eq.1	−0.926684	8.301194	1	0.0040	4.407911	4.790351	1	0.0286	13.091544	2	0.0014
Eq.2	1.011676	9.893727	1	0.0017	9.205504	93.061676	1	0.0000	102.955404	2	0.0000

Test	Statistic	df	Prob
Joint skewness	18.194921	2	0.0001
Joint kurtosis	97.852027	2	0.0000
Joint JB	116.046948	4	0.0000

[!!] WARNING: Non-normal residuals detected (joint JB $p < 0.05$). Cointegration p -values may be affected.

Johansen Cointegration Tests: Hz (no deterministic term)

TFP_unadj ~ EFF | Hz (no deterministic term) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

Unrestricted Cointegration Rank Test (Trace)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0)	0.192325	12.23736	17.95000	0.2298
At most 1 (r=1)	0.001094	0.06237	8.18000	0.9975

* Trace indicates no cointegration at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted Cointegration Rank Test (Max–Eigenvalue)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0)	0.192325	12.17498	14.90000	0.1269
At most 1 (r=1)	0.001094	0.06237	8.18000	0.9975

* Max–Eigenvalue indicates no cointegration at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

TFP_adj ~ EFF | Hz (no deterministic term) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

Unrestricted Cointegration Rank Test (Trace)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0)	0.141894	8.77052	17.95000	0.4763
At most 1 (r=1)	0.000840	0.04792	8.18000	0.9984

* Trace indicates no cointegration at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted Cointegration Rank Test (Max–Eigenvalue)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0)	0.141894	8.72260	14.90000	0.3492
At most 1 (r=1)	0.000840	0.04792	8.18000	0.9984

* Max–Eigenvalue indicates no cointegration at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Johansen Cointegration Tests: Hc (restricted constant; no trend)

TFP_unadj ~ EFF | Hc (restricted constant; no trend) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

Unrestricted Cointegration Rank Test (Trace)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.339952	31.31658	19.96000	0.0007
At most 1 (r=1)	0.125385	7.63637	9.24000	0.0959

* Trace indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted Cointegration Rank Test (Max–Eigenvalue)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.339952	23.68022	15.67000	0.0025
At most 1 (r=1)	0.125385	7.63637	9.24000	0.0959

* Max–Eigenvalue indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Normalised Cointegrating Coefficients (std. error in parentheses):

Variable	Coefficient	Std. Error	
TFP_unadj	1.000000		beta=1.0852 > 0 Valid
EFF	−1.085193	(0.10982)	
C	−0.286651	(0.13934)	

TFP_adj ~ EFF | Hc (restricted constant; no trend) | Lags (1st diff.): 1 to 2 | T=58 (1962–2019)

Unrestricted Cointegration Rank Test (Trace)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.301297	24.35786	19.96000	0.0109
At most 1 (r=1)	0.066487	3.92165	9.24000	0.3918

* Trace indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted Cointegration Rank Test (Max–Eigenvalue)

Hypothesized	Eigenvalue	Statistic	CV 5%	Prob.**
None (r=0) *	0.301297	20.43621	15.67000	0.0090
At most 1 (r=1)	0.066487	3.92165	9.24000	0.3918

* Max–Eigenvalue indicates 1 CE at 5%

** MacKinnon–Haug–Michelis (1999) p-values.

Normalised Cointegrating Coefficients (std. error in parentheses):

Variable	Coefficient	Std. Error	
TFP_adj	1.000000		beta=0.6377 > 0 Valid
EFF	−0.637697	(0.08283)	
C	−0.513240	(0.09966)	

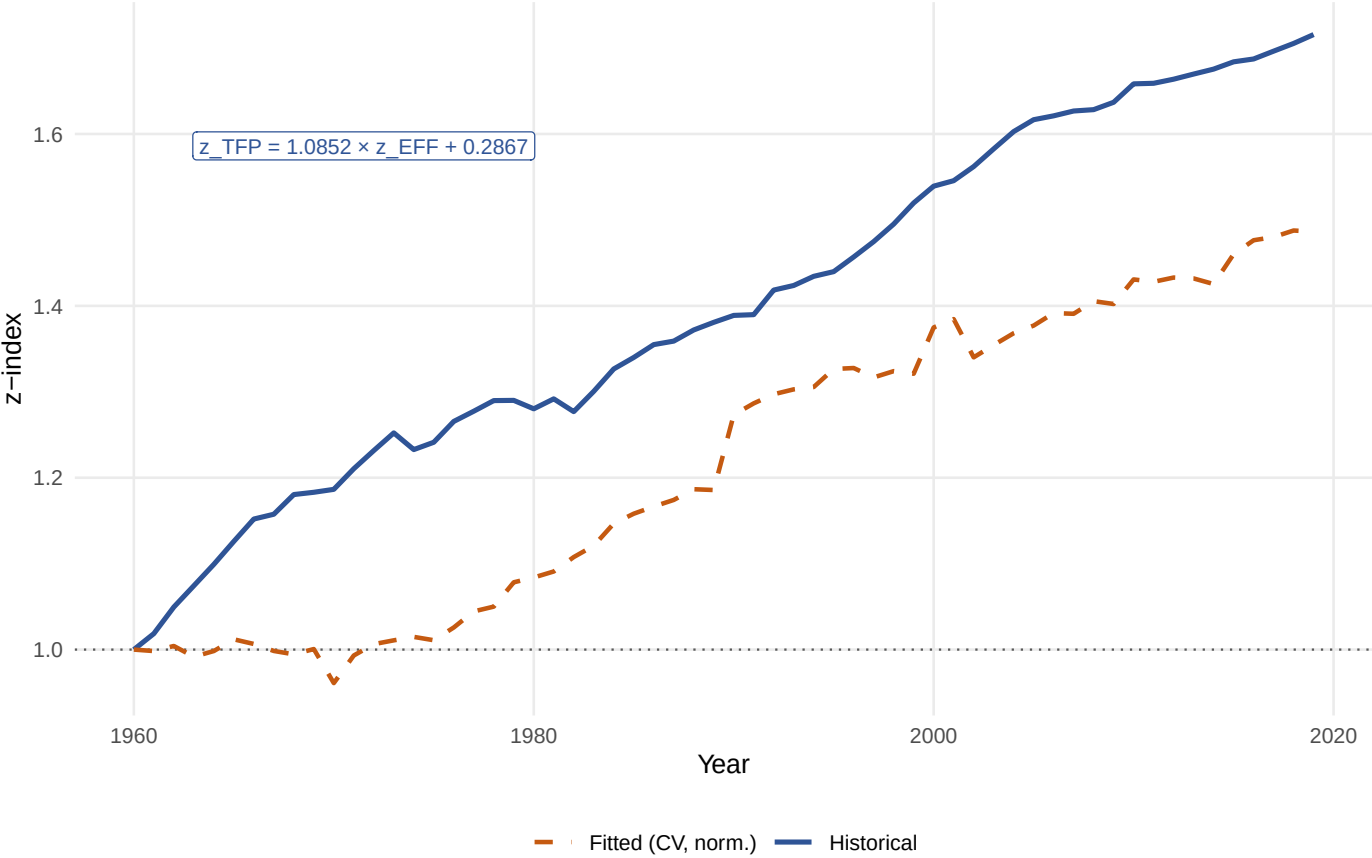
Historical vs Fitted TFP [Hz (no deterministic term)]

N/A: Unadjusted TFP

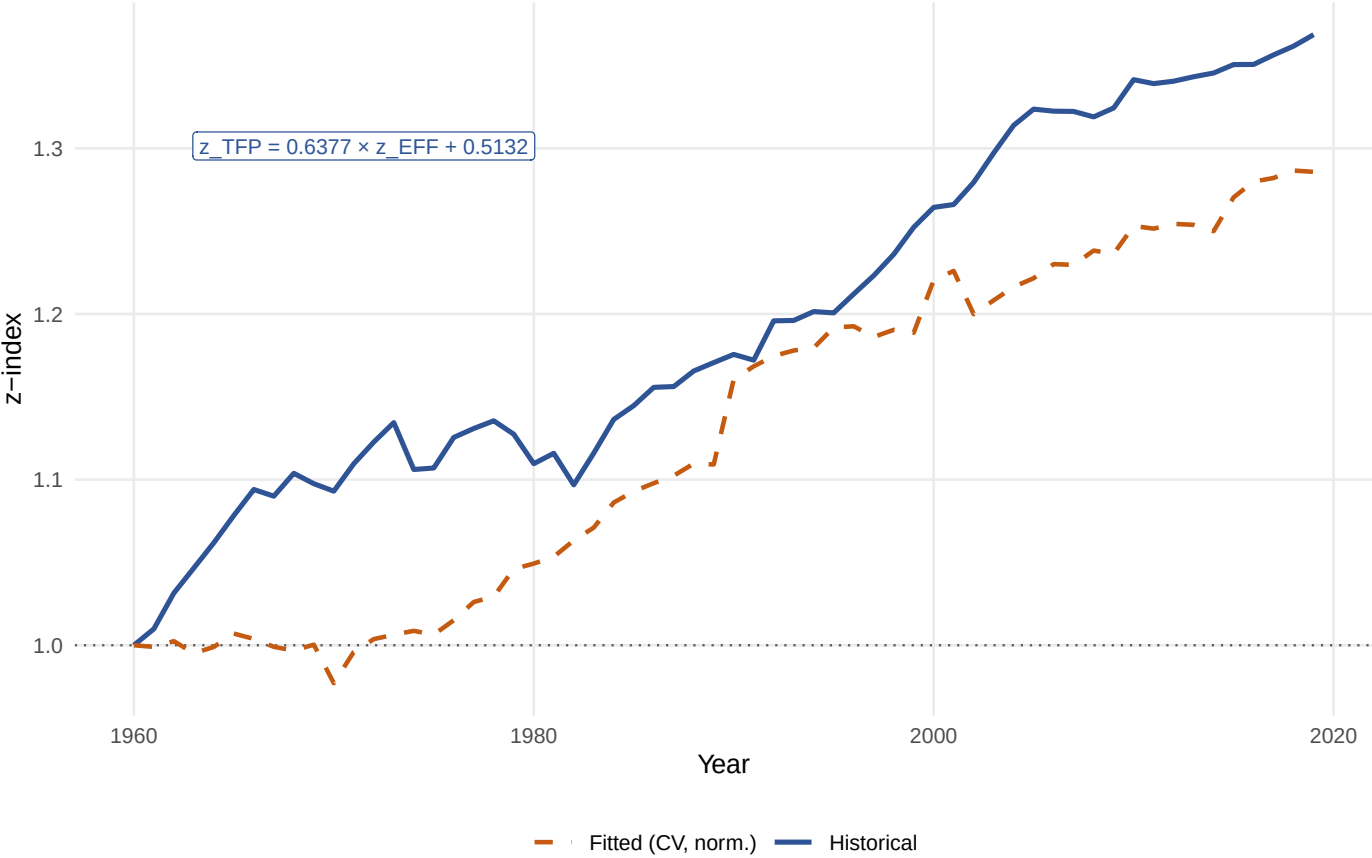
N/A: Quality-adjusted TFP

Historical vs Fitted TFP [Hc (restricted constant; no trend)]

Unadjusted TFP

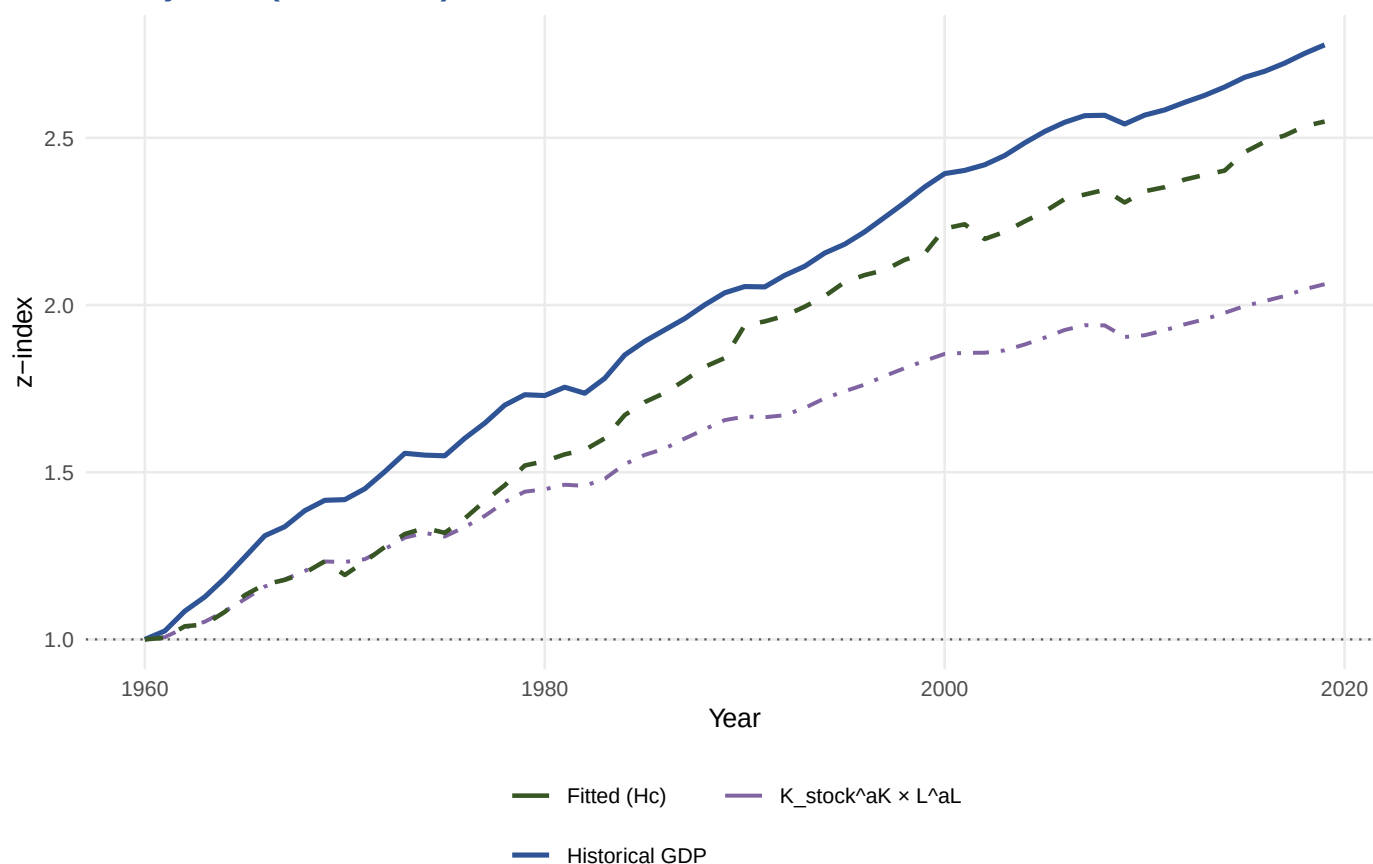


Quality-adjusted TFP

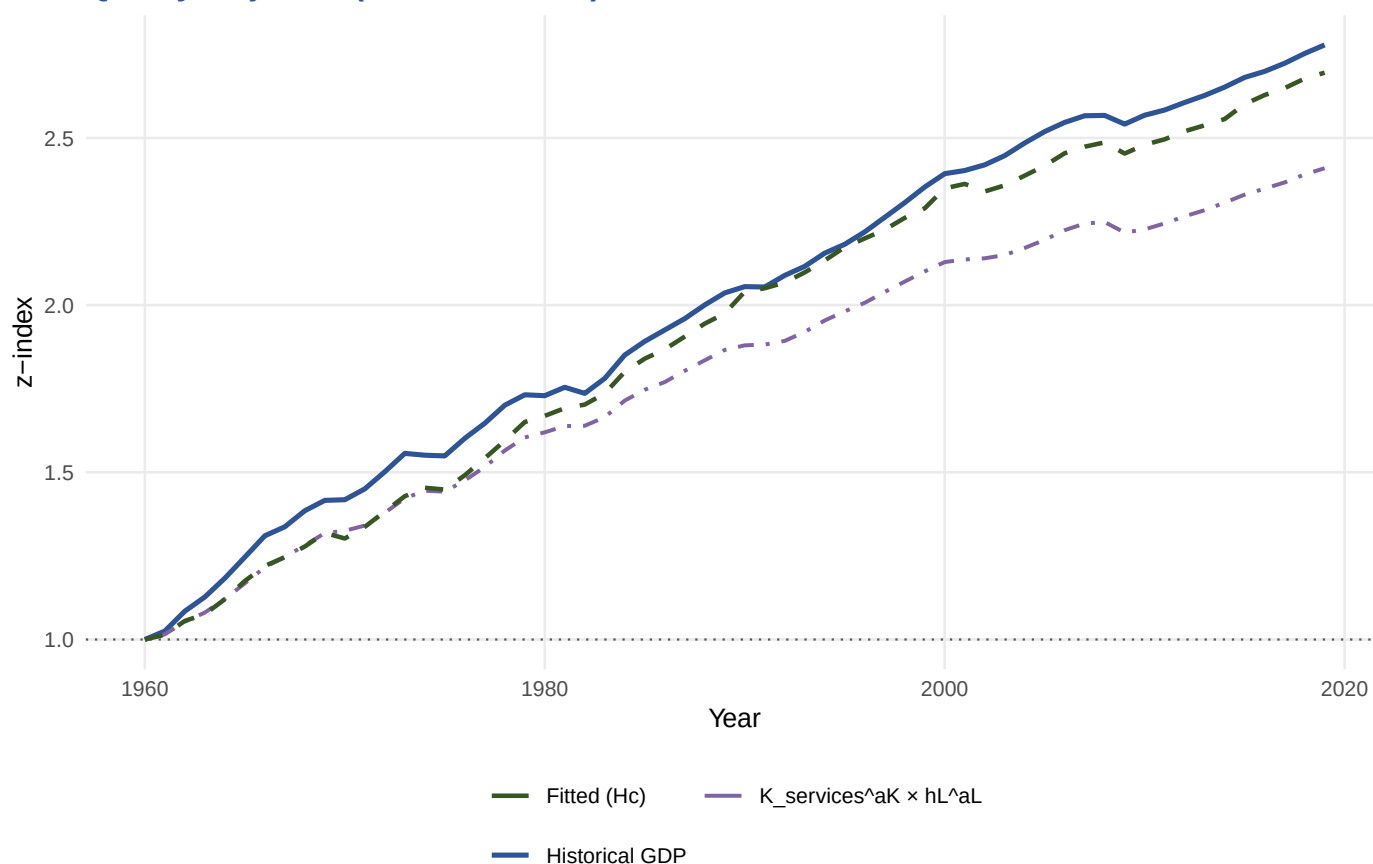


Historical vs Fitted GDP [both Hz and Hc specifications]

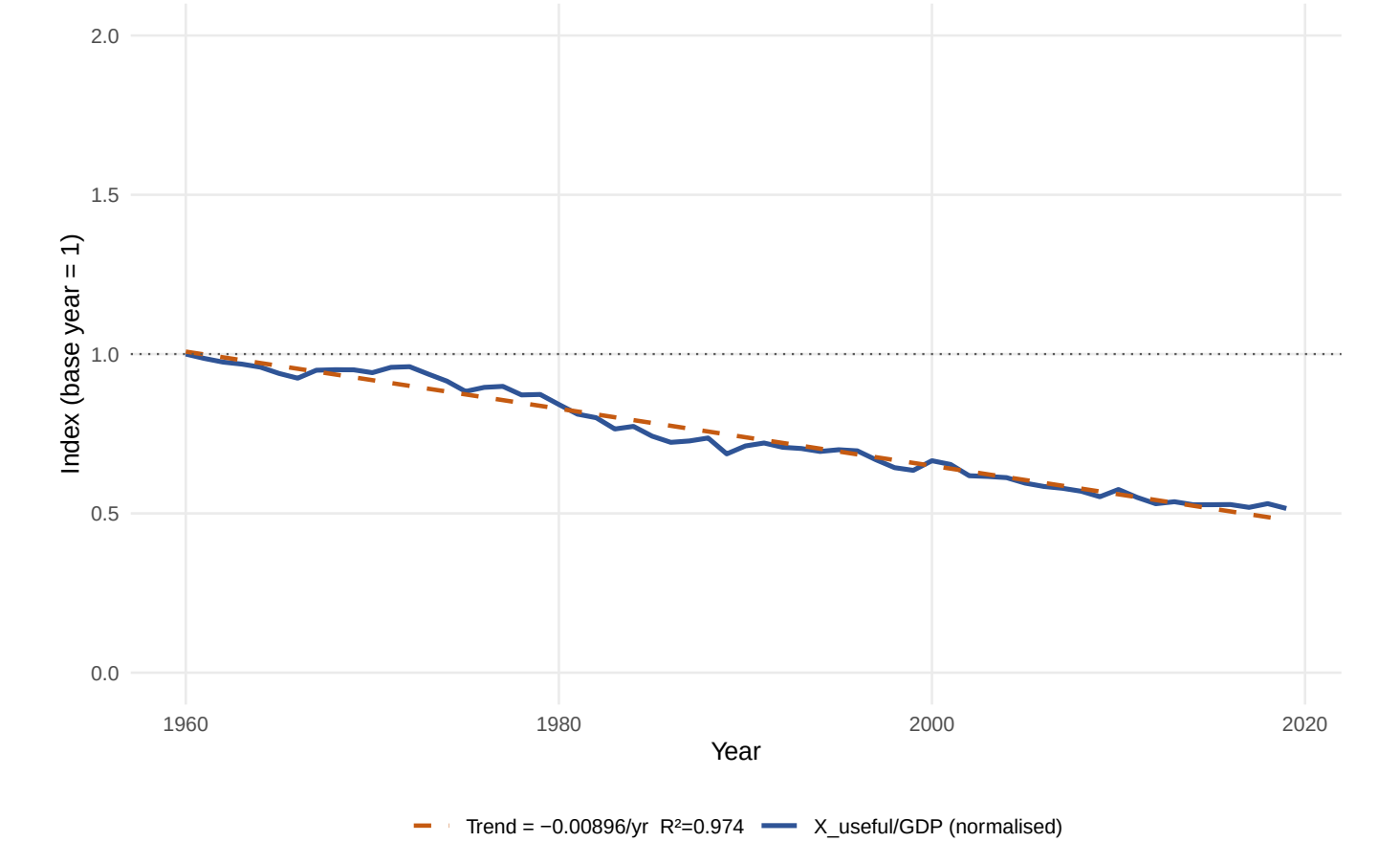
Unadjusted (K_stock, L)



Quality-adjusted (K_services, hL)



Useful Exergy Intensity [X_useful/GDP, normalised to base year = 1]



Statistic	Full (1960–2019, n=60)	Sub (2000–2019, n=20)
Mean intensity	0.8034	0.6150
Std deviation	0.1713	0.0491
CV (%)	21.32	7.98
Min	0.5570	0.5570
Max	1.0806	0.7189
OLS slope (per year)	−0.009679	−0.007906
Trend t-stat	−46.873	−13.343
Trend p-value	0.0000	0.0000
Trend R2	0.9743	0.9082
Annual growth rate (% p.a.)	−1.205	−1.285
ADF statistic (levels)	−2.3772	−0.9604
ADF CV 5%	−3.4425	−3.1447
PP statistic (levels)	−2.2958	−2.5031
PP CV 5%	−3.4862	−3.6746

Individual test results

Statistic	Full (1960–2019, n=60)	Sub (2000–2019, n=20)
No significant trend (p>0.10)	No	No
ADF: I(0) at 5%	No	No
PP: I(0) at 5%	No	No
CV < 5%	No	No
VERDICT: Approx. constant?	No	No

References

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